Bachelor's Thesis Timeline

10-Week Plan · Benchmarking Tabular Data Augmentation Techniques for Deep Clustering

Hai Nguyen · B.Sc. Computer Science, LMU Munich

Topic	Benchmarking Tabular Data Augmentation Techniques for Deep Clustering
Official processing time	16 June 2026 – 25 August 2026 (10 weeks)
Preparation phase	until 15 June 2026 — before the official start
Weekly workload	approx. 20–25 hours (part-time)
Estimated total effort	approx. 220 hours (official phase)
Submission deadline	Monday, 25 August 2026 at the latest
Prior knowledge	Data Mining course (clustering, NMI/ARI, silhouette); Python tooling and PyTorch are learned during the preparation phase

1 Overview

This plan divides the bachelor's thesis into a preparation phase and five phases across ten official weeks. The central challenge: the topic requires PyTorch and deep learning, while I am starting without experience in numpy, pandas, matplotlib and scikit-learn. Because the official processing time is only ten weeks, all tool onboarding is moved into the preparation phase before 15 June — this keeps the full official period available for the actual research work.

My Data Mining course is a real asset here: clustering methods and the evaluation metrics NMI, ARI and silhouette are already familiar to me. The learning load therefore clearly sits in two areas — the Python tooling and the deep learning part (contrastive learning, PyTorch).

The plan follows three principles: the tools are learned before the official start, a minimal end-to-end pipeline is brought to a working state early (week 4), and writing happens continuously rather than only at the end.

2 Preparation phase (until 15 June)

The weeks before the official start on 16 June are firmly reserved for tool onboarding. The goal is to begin on 16 June with a working environment and solid Python skills — so that none of the ten official weeks is spent on pure tool learning.

Tasks

- Set up the Python environment: Miniconda or venv, Jupyter Notebook or VS Code, create a Git repository
- numpy & pandas: arrays and DataFrames, indexing, loading CSV files, simple transformations
- matplotlib: scatter plots, histograms and learning curves
- scikit-learn: StandardScaler, KMeans, plus the metrics NMI, ARI and silhouette
- Mini-project: cluster the Iris or Digits dataset, compute NMI/ARI, plot the embedding in 2D — directly builds on your Data Mining course
- Getting started with PyTorch: tensors and autograd via the tutorial “Deep Learning with PyTorch: A 60 Minute Blitz”
- Skim SCARF (Bahri et al., 2022) once to get a feel for the topic

Result: A working environment, Python fundamentals, and an initial feel for PyTorch and the topic.

3 Phase overview

Phase	Weeks	Period	Focus	Effort
0	–	until 15 Jun	Preparation: tools & environment	lead-in
A	1–2	15–28 Jun	Literature review & conception	≈ 44 h
B	3–5	29 Jun–19 Jul	Implementing the benchmark	≈ 66 h
C	6–7	20 Jul–2 Aug	Experiments & sensitivity studies	≈ 44 h
D	8–9	3–16 Aug	Analysis & writing	≈ 44 h
E	10	17–23 Aug	Revision & submission	≈ 22 h
			Total effort, official phase (estimate)	**≈ 220 h**

4 Detailed weekly plan

PHASE A · LITERATURE REVIEW & CONCEPTION

Week 1 · 15–22 Jun 2026

Literature review & consolidating PyTorch

Tasks

- Read SCARF (Bahri et al., 2022) thoroughly — the central methodological reference for feature corruption and contrastive learning on tabular data
- Skim the augmentation survey (Wang et al., 2025), focusing on the sections on tabular data
- Review 3–4 further sources: self-supervised/contrastive learning for tabular data (e.g. VIME, SubTab), deep clustering fundamentals (DEC/IDEC), the SimCLR principle
- Structure the literature notes and maintain bib-refs.bib (BibTeX)
- Consolidate PyTorch: nn.Module, Dataset/DataLoader, training loop and optimizer — building on the preparation phase
- Exercise: train a simple MLP (autoencoder or classifier) on a tabular dataset

Tools & learning focus: PyTorch (full training pipeline), reference management with BibTeX.

Result: Solid PyTorch skills and a well-founded literature overview. ◆ **Milestone M1**

Week 2 · 22–29 Jun 2026

Fixing the method & drafting Related Work

Tasks

- Fix the methodology: contrastive pretraining of an MLP encoder followed by k-means on the embeddings as the deep clustering backbone
- Select datasets: 3–4 tabular datasets with ground-truth labels (OpenML/UCI), varying in size and dimensionality
- Classify each selected dataset by feature type: compute the fraction of categorical vs. numerical features, and (where feasible) identify which features carry the most mutual information with the cluster labels — this taxonomy drives stratified reporting in Phase C
- Define the thesis structure; write a rough draft of the “Related Work” chapter
- Supervisor meeting: agree on method, datasets and scope

Tools & learning focus: LaTeX (cleanthesis template), literature work.

Result: Method and datasets fixed, related-work draft, scope confirmed with the supervisor. ◆ **Milestone M2**

PHASE B · IMPLEMENTING THE BENCHMARK

Week 3 · 29 Jun–5 Jul 2026

Data pipeline & encoder

Tasks

- Data module: loading, preprocessing (scaling numerical and encoding categorical features), train/test split
- Augmentation module as a scaffold with the first technique: feature masking / corruption in the SCARF style
- Implement a feature-type mask in the augmentation module: tag each column as categorical or numerical; numerical features receive Gaussian noise or feature-wise scaling, categorical features are either left unchanged or replaced with values drawn from that feature's empirical distribution — never arbitrary corruption
- Implement the encoder network (MLP) that outputs embeddings

Tools & learning focus: PyTorch, pandas.

Result: Data is loaded and augmented correctly; the encoder produces embeddings.

Week 4 · 6–13 Jul 2026

Contrastive training & first end-to-end pipeline

Tasks

- Implement the contrastive loss (NT-Xent / InfoNCE)
- Assemble the training loop and connect it to the encoder
- First full run: one dataset, one augmentation → embedding → k-means → NMI/ARI
- Debugging and sanity checks: does the loss decrease, are the embeddings meaningfully separated?

Tools & learning focus: PyTorch.

Result: A minimal working pipeline (MVP) produces first clustering results. ◆ **Milestone M3**

Week 5 · 13–19 Jul 2026

Completing the benchmark

Tasks

- Add further augmentations: Gaussian noise, feature swapping/shuffling, optionally zeroing or mixup
- Add no-augmentation baseline: train the contrastive model with identity augmentation (each row passed twice, unchanged) — this is the theoretical upper bound for semantic preservation; label it "Identity" in all result tables
- Add feature-type-aware pipeline as a fourth named technique: semantically conservative augmentation using the feature-type mask from Week 3; label it "Type-Aware" in all result tables alongside the three main techniques
- Ensure reproducibility: fixed seeds, configuration files, log results as CSV
- Extend the code to all selected datasets and clean it up
- Plan buffer time for debugging

Tools & learning focus: PyTorch, Git.

Result: A complete, reproducible benchmark codebase.

PHASE C · EXPERIMENTS & SENSITIVITY STUDIES

Week 6 · 20–26 Jul 2026

Main experiments

Tasks

- Run the full experiments: all augmentations × all datasets, several seeds each
- Use Google Colab for GPU acceleration if needed

- Store results systematically and traceably

Tools & learning focus: PyTorch, Google Colab.

Result: Raw results for all main configurations are available.

Week 7 · 27 Jul–3 Aug 2026

Sensitivity studies & failure cases

Tasks

- Hyperparameter and sensitivity studies: masking rate, augmentation strength, embedding dimension
- Investigate limitations and failure cases of the augmentation techniques
- Produce stratified results tables: report NMI and ARI broken down by dataset taxonomy (e.g. datasets with >50% categorical features vs. mostly numerical) — this turns the semantic-preservation concern into a testable, clearly structured hypothesis
- Prepare results as tables and figures
- Supervisor meeting: discuss the results

Tools & learning focus: PyTorch, matplotlib.

Result: Complete experimental results incl. plots and tables. ◆ **Milestone M4**

PHASE D · ANALYSIS & WRITING

Week 8 · 3–9 Aug 2026

Analysis & core chapters

Tasks

- Analyse the results: which augmentation improves or worsens clustering — and possible explanations
- Analyse stratified results: does the no-augmentation baseline match or outperform augmented variants on high-categorical datasets? Does the Type-Aware pipeline improve clustering on those datasets? Report findings per taxonomy stratum
- Write the "Methodology" chapter (LaTeX)
- Write the "Experiments & Results" chapter

Tools & learning focus: LaTeX, matplotlib.

Result: Analysis completed; methodology and results chapters drafted.

Week 9 · 10–17 Aug 2026

Discussion & complete first draft

Tasks

- Write the "Discussion" chapter: interpretation, contextualisation, limitations
- Write the introduction, related work (finalise), conclusion & outlook, and abstract
- Merge the full document and hand it to the supervisor

Tools & learning focus: LaTeX.

Result: Complete first draft handed to the supervisor. ◆ **Milestone M5**

PHASE E · REVISION & SUBMISSION

Week 10 · 17–25 Aug 2026

Revision & submission

Tasks

- Incorporate supervisor feedback
- Proofread: language, consistency, narrative flow

- LaTeX polishing: formatting, references with biber, lists/indexes, figure quality
- Declaration of authorship, final checks, submission format
- Buffer for unforeseen issues

Tools & learning focus: LaTeX.

Result: Submission of the bachelor's thesis — Sunday, 25 Aug 2026 at the latest. **◆ Milestone M6**

5 Milestones

Six milestones make progress verifiable. If a milestone is clearly missed, that is a strong signal to talk to the supervisor about adjusting the scope.

	When	Date	Outcome
M1	End of week 1	22 Jun 2026	PyTorch fundamentals consolidated; well-founded literature overview
M2	End of week 2	29 Jun 2026	Method and datasets fixed; related-work draft; scope agreed with the supervisor
M3	End of week 4	13 Jul 2026	Minimal end-to-end pipeline (MVP) produces first clustering results
M4	End of week 7	3 Aug 2026	Complete experimental results incl. sensitivity studies, tables and figures
M5	End of week 9	17 Aug 2026	Complete first draft handed to the supervisor
M6	End of week 10	25 Aug 2026	Submission of the bachelor's thesis (latest date)